

Master's Thesis, Goethe University Frankfurt, 2025
Portfolio Optimization Using Machine Learning

The most interesting quantitative investment project I've done was my master's thesis, where I built an end-to-end systematic portfolio strategy that bridges machine-learning return forecasts with downside risk optimization framework. The goal was simple: test whether modern ML models (like Support Vector Regression, SVR, and Random Forest, RF) can improve portfolio performance versus a traditional baseline (ARIMA) when their forecasts are fed into a Conditional Value-at-Risk (CVaR) allocation engine.

Data

I worked with a 12-year weekly dataset (Jan-2013 to Jan-2025; 600 + observations) covering seven liquid futures across equities, rates, and commodities: ES1 (S&P 500), VG1 (Euro Stoxx 50), Z1 (FTSE 100), RX1 (Euro-Bund), GC1 (Gold), CO1 (Brent), and HG1 (Copper). To power the ML models, I created a rich feature set spanning macro and market indicators (e.g. CDS indices, VIX/V2X, CBOE SKEW Index, DXY Curncy, MSCI ACWI Index, sector ETFs like XLF/XLE, LMEX, EMBI), plus technical signals (EMA, MACD/Signal, Bollinger Bands, RSI, ROC, historical volatility). Most inputs were transformed to weekly frequency and log changes where appropriate.

Modeling Approach

I implemented two evaluation regimes: (i) a single train/test split (train: Apr-2013 to Dec-2018; test: Jan-2019 to Jan-2025) and (ii) a rolling-window workflow. Each model predicted one step ahead and was retrained every 24 out-of-sample observations. To reduce leakage and stabilize generalization, I used TimeSeriesSplit cross-validation and Bayesian hyperparameter optimization (BayesSearchCV). Features were scaled (SVR needs it; it also helped RF), and I applied adaptive feature selection via mutual information, tuned per asset (and fixed sensibly in the rolling setup to control compute).

Optimization & Portfolio Construction

Forecasted weekly returns served as expected returns in a CVaR optimization at the 90% VaR threshold, using historical scenarios (non-parametric). I imposed realistic constraints - no shorting, 40% cap per asset - to avoid concentration, rebalanced biweekly, and charged 1 bps per traded notional to account for transaction costs. Benchmarks were an equal-weight (EW) futures portfolio and the S&P 500. I evaluated both forecasting quality (MSE, R^2 , Directional Accuracy, Spearman rank) and portfolio outcomes (annualized Sharpe, Max Drawdown, cumulative return).

Results

ARIMA underperformed as a static forecaster and degraded further in rolling mode (Sharpe¹ 0.12; Cumulative Ret 1.05), confirming its limits in non-stationary regimes. SVR's baseline already produced strong Directional Accuracy and rank ordering; with rolling retraining it improved further (Sharpe 0.64; Cumulative Ret 1.6), showing that correct ranking matters in a CVaR framework even when R^2 is modest. RF excelled once retrained regularly: RF-Rolling achieved the best risk-adjusted profile (Sharpe 0.75; MaxDD 16.0%; Cumulative Ret 1.7), showing performance as smooth as the S&P 500, although with lower total returns (Sharpe 0.85; MaxDD 31.8%; Cumulative Ret 2.3). Notably, ML portfolios held up better during regime shifts (COVID-19 shock, 2022 volatility) and accelerated as markets normalized, highlighting the value of frequent updates. An insightful by-product was interpretability. RF feature importance in rolling mode aligned with market intuition: global equities (MXWD) dominated equity futures; energy (XLE) mattered for oil; LMEX for copper; DXY and bond/volatility factors for gold; and rate/bond indices for Bunds. This gave the strategy "economic explainability".

Learning outcomes

Three practical takeaways stood out. First, updating beats cleverness: rolling retraining outperformed static gains. Second, ranking outperformed point accuracy under CVaR-Directional Accuracy and Spearman often mattered more than R^2 . Third, risk modeling must reflect investor loss aversion: CVaR paired naturally with ML forecasts to emphasize asymmetric downside control.

Limitations & Next Steps

I modeled simple trading frictions (no slippage/market impact) and used weekly data, which may miss shorter-horizon alpha. Future work may include incorporating intraday features, more granular cost modeling, and the application of deep learning architectures, provided sufficient computational resources are available.

¹Sharpe ratio was annualized in all cases